



RGA

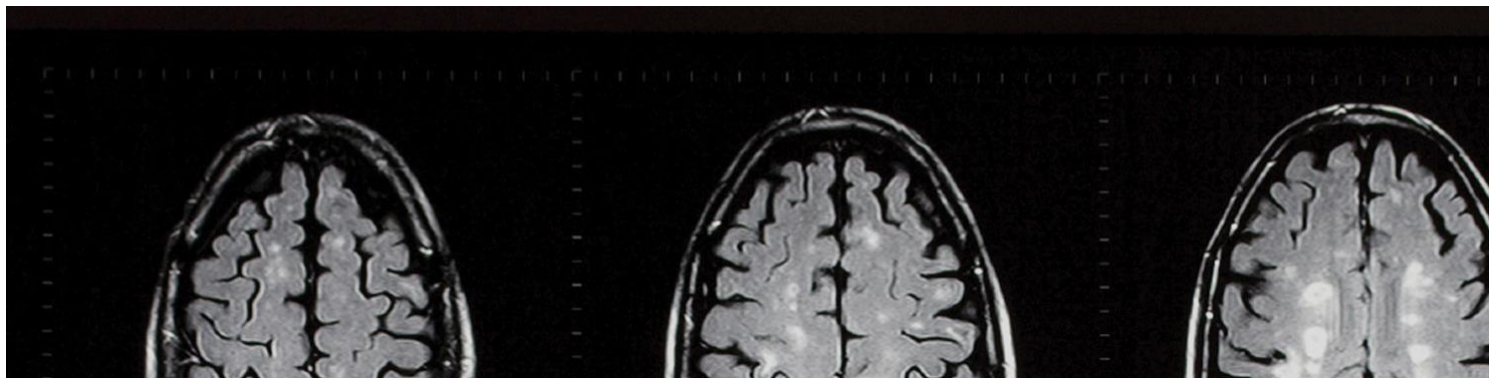
Multiple Sclerosis

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VP, Head of Underwriting Manual Development

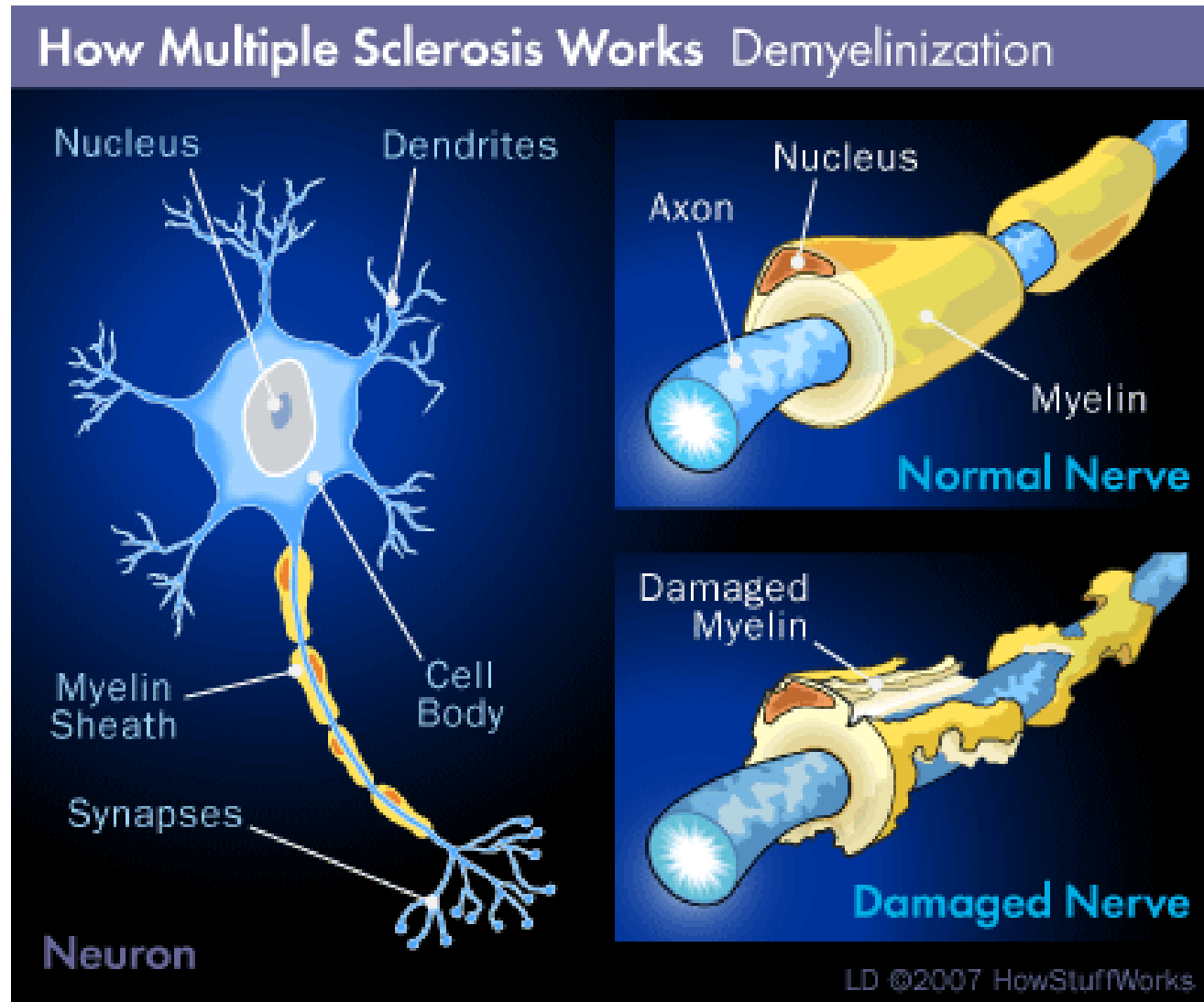
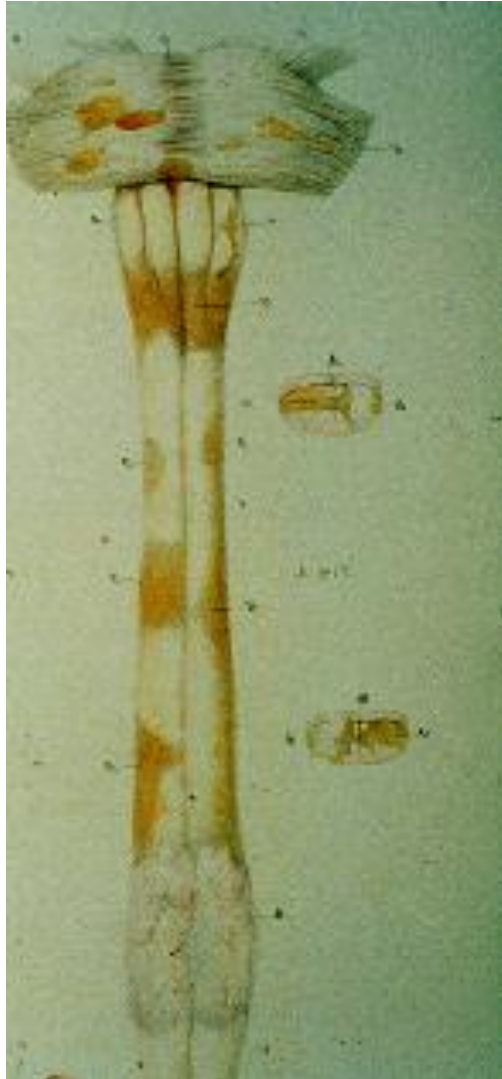
September 2016

Agenda



- What is Multiple Sclerosis?
- How does it affect the body?
- How is it diagnosed?
- What's the pattern of disease?
- Treatment?
- Evidence requirements.....

Natural history of Multiple Sclerosis



Source: <http://health.howstuffworks.com/multiple-sclerosis1.htm>

Diversity of symptoms

Main symptoms of Multiple sclerosis

Central:

- Fatigue
- Cognitive impairment
- Depression
- Unstable mood

Visual:

- Nystagmus
- Optic neuritis
- Diplopia

Speech:

- Dysarthria

Throat:

- Dysphagia

Musculoskeletal:

- Weakness
- Spasms
- Ataxia

Sensation:

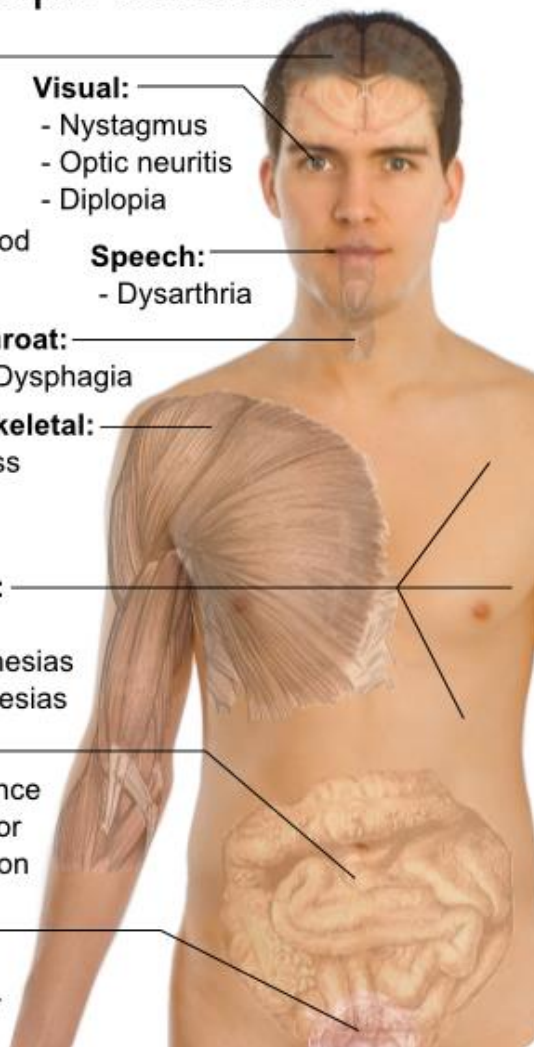
- Pain
- Hypoesthesias
- Paraesthesias

Bowel:

- Incontinence
- Diarrhea or constipation

Urinary:

- Incontinence
- Frequency or retention



Source: [http://www.news-medical.net/health/Multiple-Sclerosis-\(MS\).aspx](http://www.news-medical.net/health/Multiple-Sclerosis-(MS).aspx)

Multiple sclerosis symptoms

Clinical features of multiple sclerosis

Features suggestive of multiple sclerosis
Relapses and remissions
Onset between ages 15 and 50
Optic neuritis
Lhermitte's sign
Internuclear ophthalmoplegia
Fatigue
Uhthoff's phenomenon
Features not suggestive of multiple sclerosis
Steady progression
Onset before age 10 or after age 50
Cortical deficits such as aphasia, apraxia, alexia, neglect
Rigidity, sustained dystonia
Convulsions
Early dementia
Deficit developing within minutes



Multiple Sclerosis Symptoms

Suggestive of MS

- Relapses and remissions.
- Onset between age 15-50 years.
- Optic neuritis.
- L'hermitte's sign.
- Uthoff's phenomenon.
- Fatigue.

Not suggestive of MS

- Steady progression.
- Onset before 10 or after 50.
- Cortical deficits such as aphasia, apraxia.
- Rigidity, sustained dystonia.
- Convulsions.
- Early dementia.
- Neurological deficit developing within minutes.

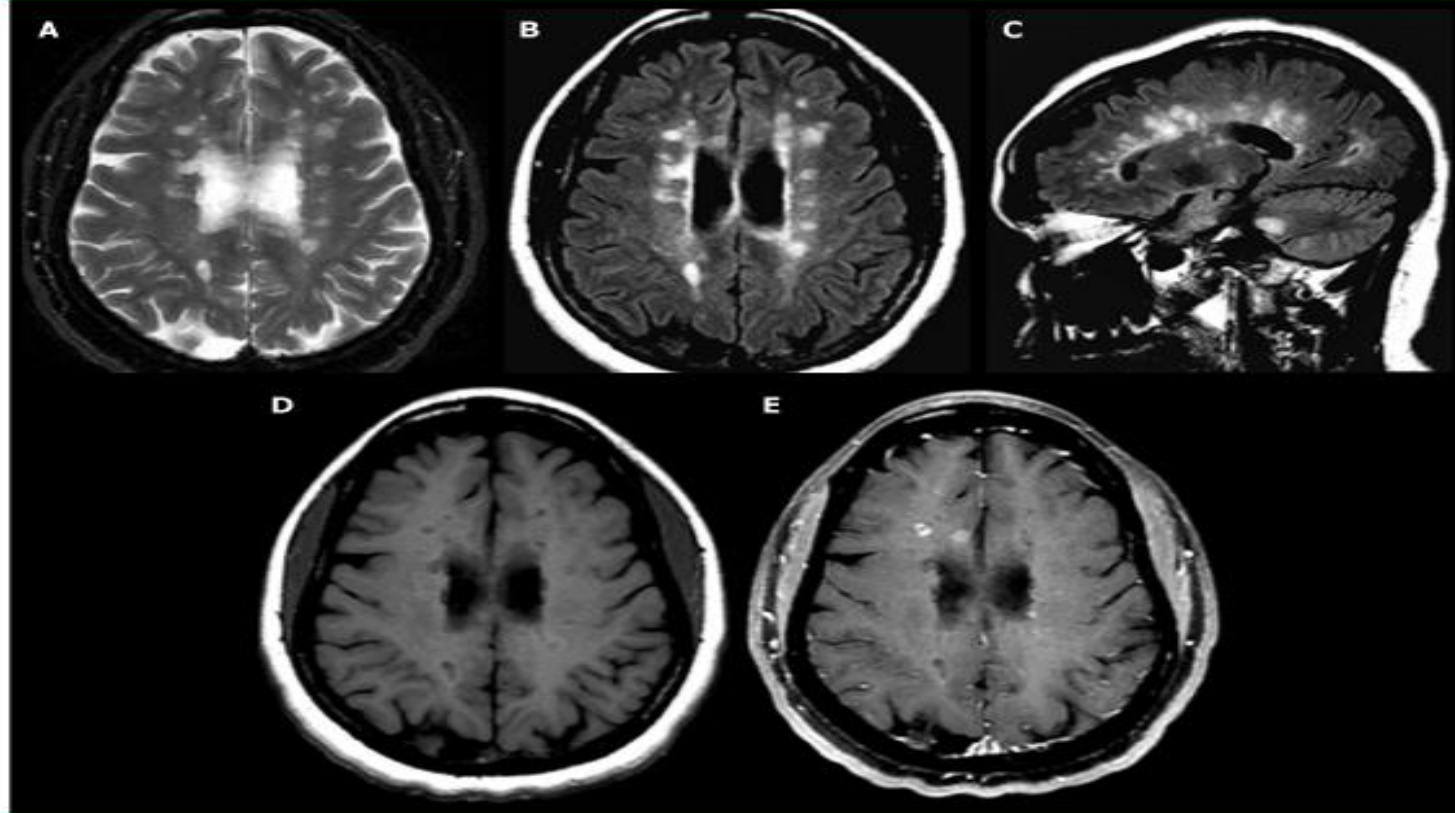
Diagnosis...

- Dissemination in time and space....
- McDonald Criteria....
- Diagnostic tools
 - MRI
 - Lumbar Puncture
 - Visual Evoked Responses

2010 McDonald Criteria.

Clinical Presentation	Additional Data Needed for MS Diagnosis
≥ 2 attacks ^a ; objective clinical evidence of ≥ 2 lesions or objective clinical evidence of 1 lesion with reasonable historical evidence of a prior attack ^b	None ^c
≥ 2 attacks ^a ; objective clinical evidence of 1 lesion	Dissemination in space, demonstrated by: ≥ 1 T2 lesion in at least 2 of 4 MS-typical regions of the CNS (periventricular, juxtacortical, infratentorial, or spinal cord) ^d ; or Await a further clinical attack ^a implicating a different CNS site
1 attack ^a ; objective clinical evidence of ≥ 2 lesions	Dissemination in time, demonstrated by: Simultaneous presence of asymptomatic gadolinium-enhancing and nonenhancing lesions at any time; or A new T2 and/or gadolinium-enhancing lesion(s) on follow-up MRI, irrespective of its timing with reference to a baseline scan; or Await a second clinical attack ^a
1 attack ^a ; objective clinical evidence of 1 lesion (clinically isolated syndrome)	Dissemination in space and time, demonstrated by: For DIS: ≥ 1 T2 lesion in at least 2 of 4 MS-typical regions of the CNS (periventricular, juxtacortical, infratentorial, or spinal cord) ^d ; or Await a second clinical attack ^a implicating a different CNS site; and For DIT: Simultaneous presence of asymptomatic gadolinium-enhancing and nonenhancing lesions at any time; or A new T2 and/or gadolinium-enhancing lesion(s) on follow-up MRI, irrespective of its timing with reference to a baseline scan; or Await a second clinical attack ^a
Insidious neurological progression suggestive of MS (PPMS)	1 year of disease progression (retrospectively or prospectively determined) plus 2 of 3 of the following criteria ^d : 1. Evidence for DIS in the brain based on ≥ 1 T2 lesions in the MS-characteristic (periventricular, juxtacortical, or infratentorial) regions 2. Evidence for DIS in the spinal cord based on ≥ 2 T2 lesions in the cord 3. Positive CSF (isoelectric focusing evidence of oligoclonal bands and/or elevated IgG index)

Brain MRI of a 42 year-old woman with multiple sclerosis

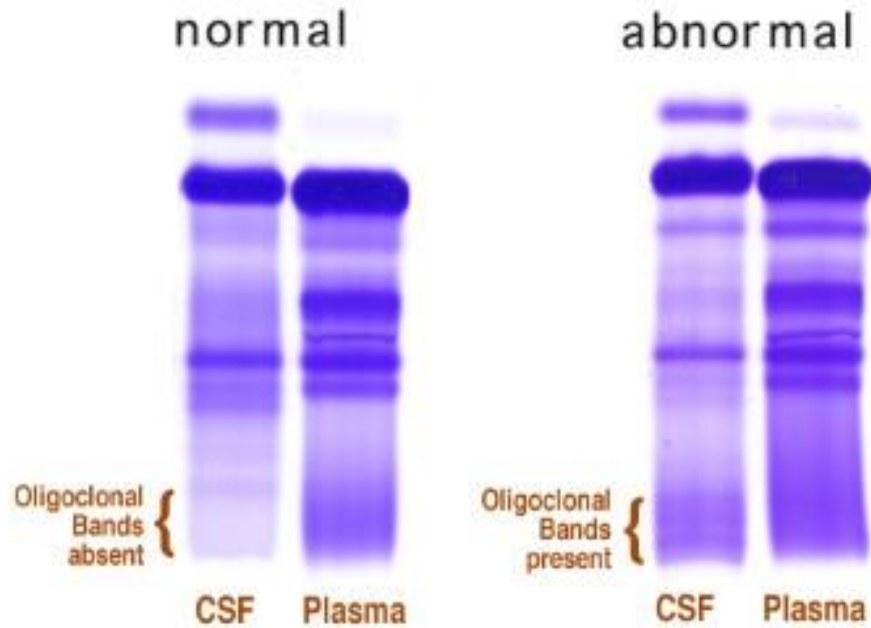


Axial T2-weighted (A) and axial fluid attenuated inversion recovery (FLAIR) (B) images show multiple, ovoid shaped, hyperintense foci consistent with multiple sclerosis plaques. Sagittal FLAIR (C) image also shows these lesions to be radiating out from the corpus callosum. Axial precontrast T1-weighted (D) image shows that many of these lesions are hypointense, consistent with black holes. Axial postgadolinium fat saturated T1-weighted (E) image shows that some of these plaques enhance in a ring-like fashion consistent with active plaques.



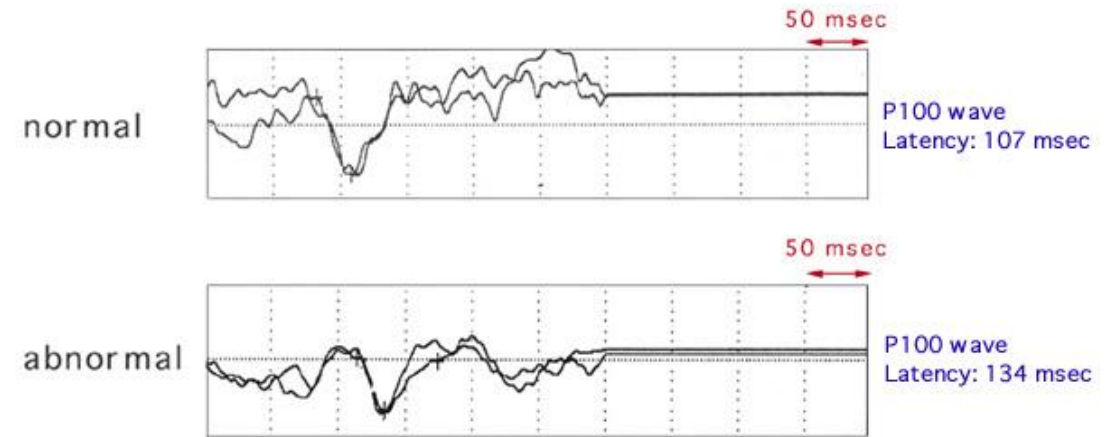
<http://img.medscape.com/pi/features/slideshow-slide/multiple-sclerosis/fig10.jpg>

Oligoclonal Bands in CSF



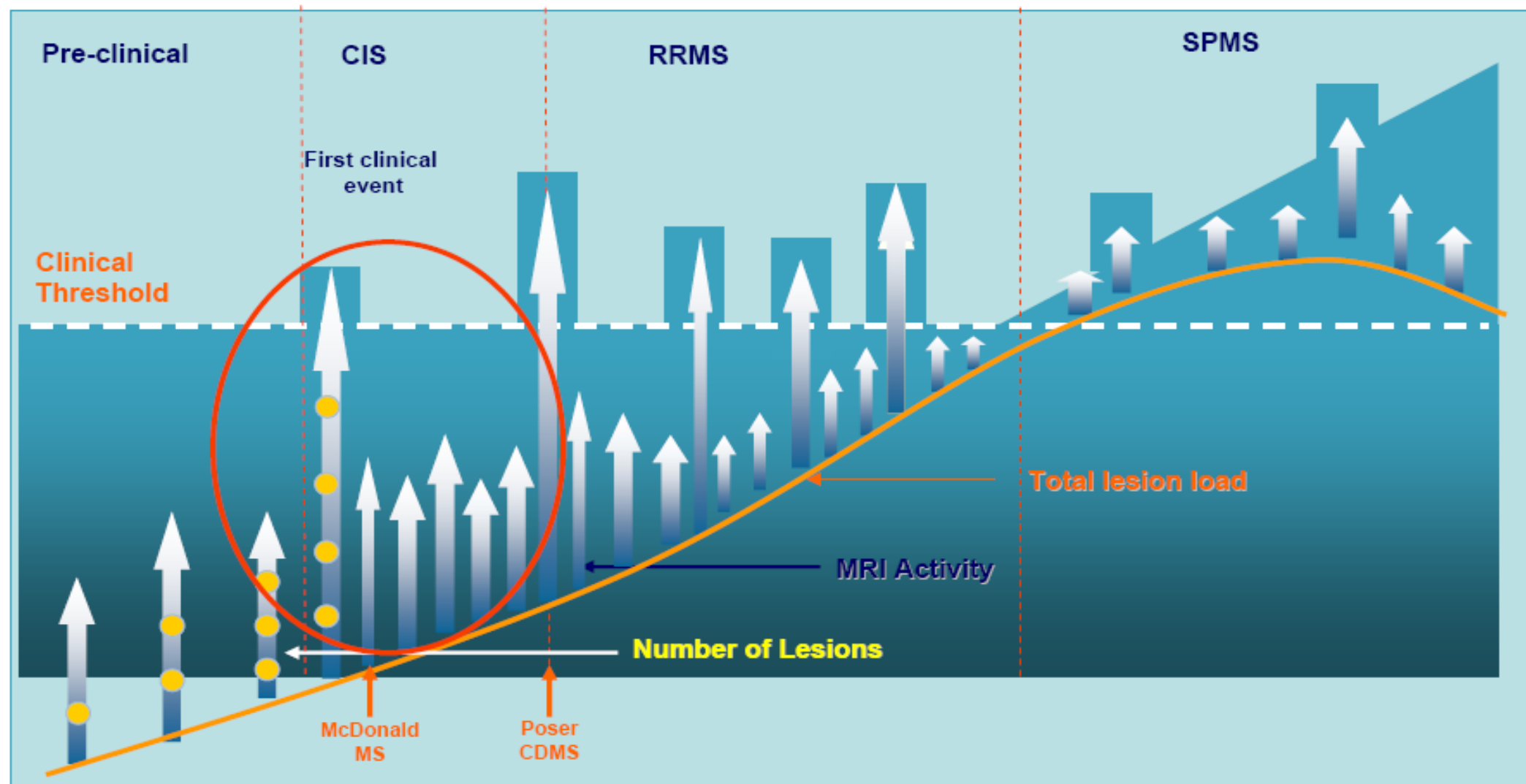
<http://www.newportdoctorsmedicalgroup.com/images/csf.jpg>

Visual Evoked Potentials



http://library.med.utah.edu/kw/ms/mml/ms_vep.jpg

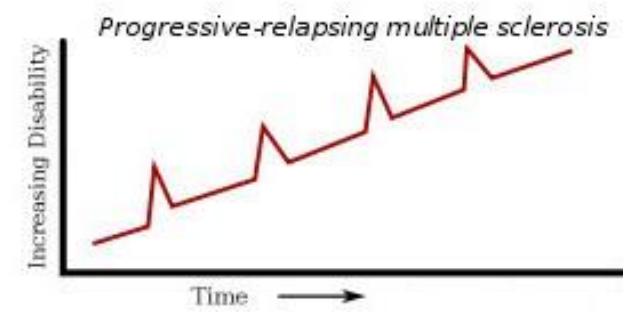
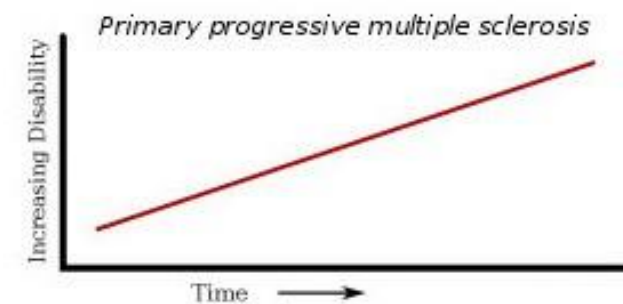
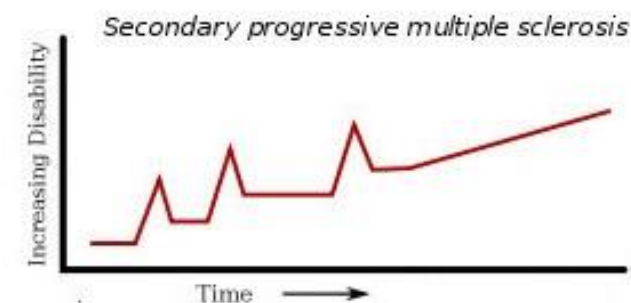
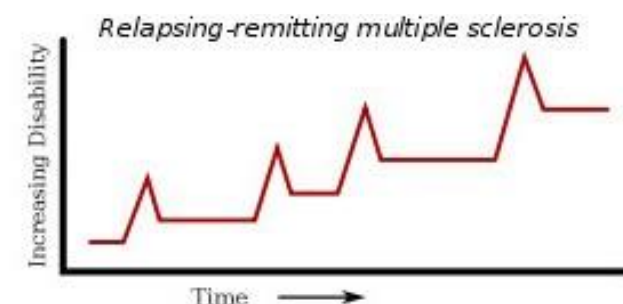
Clinical progression of MS....



Source: Recent insights into the management of MS. Omar Malik, Consultant Neurologist Imperial College Healthcare NHS Trust. Courtesy of MS Society, UK

Pattern of disease....

- Benign?
- Relapsing Remitting.
- Secondary Progressive.
- Primary Progressive.
- Progressive Relapsing.



Treatment

- Infusion
- Injection
- Oral
- Others



Treatment...

Injection

- Interferons
 - Interferon beta – 1b (Betaseron)
 - Interferon beta – 1a (Avonex/Rebif)
 - Long term efficacy?
 - Side effects.

- Glatiramer Acetate



Treatment

Infusion and Oral....

- Infusion:
 - Natalizumab (Tysabri)
 - Alemtuzumab
 - Mitoxantrone

- Oral
 - Dimethyl fumarate.
 - Teriflunomide.
 - Fingolimod



Treatment...

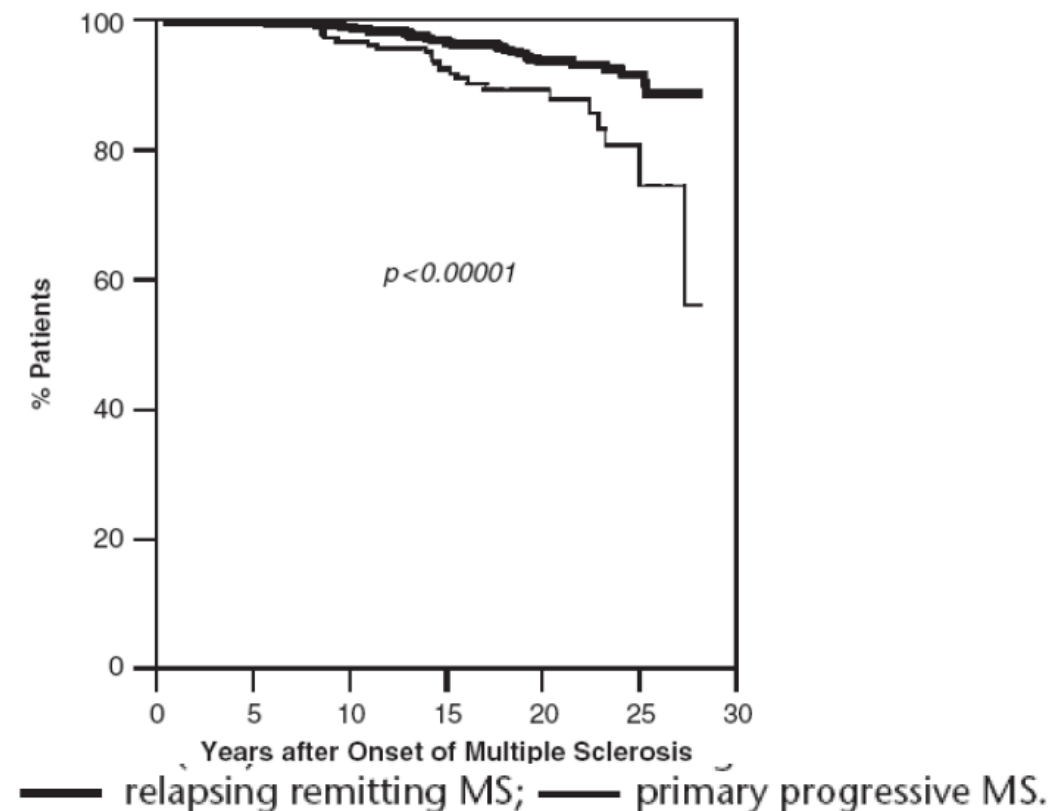
Others

- Azathioprine.
- Cyclophosphamide.
- Glucocorticoids.



Prognostic factors

- Type of disease – RRMS vs. PPMS.
- Early disease course?
- Lesion load.
- Pregnancy.
- Psychosocial stress.
- Time to EDSS 3....



Kaplan-Meier analyses of the time from multiple sclerosis onset to death according to initial course.

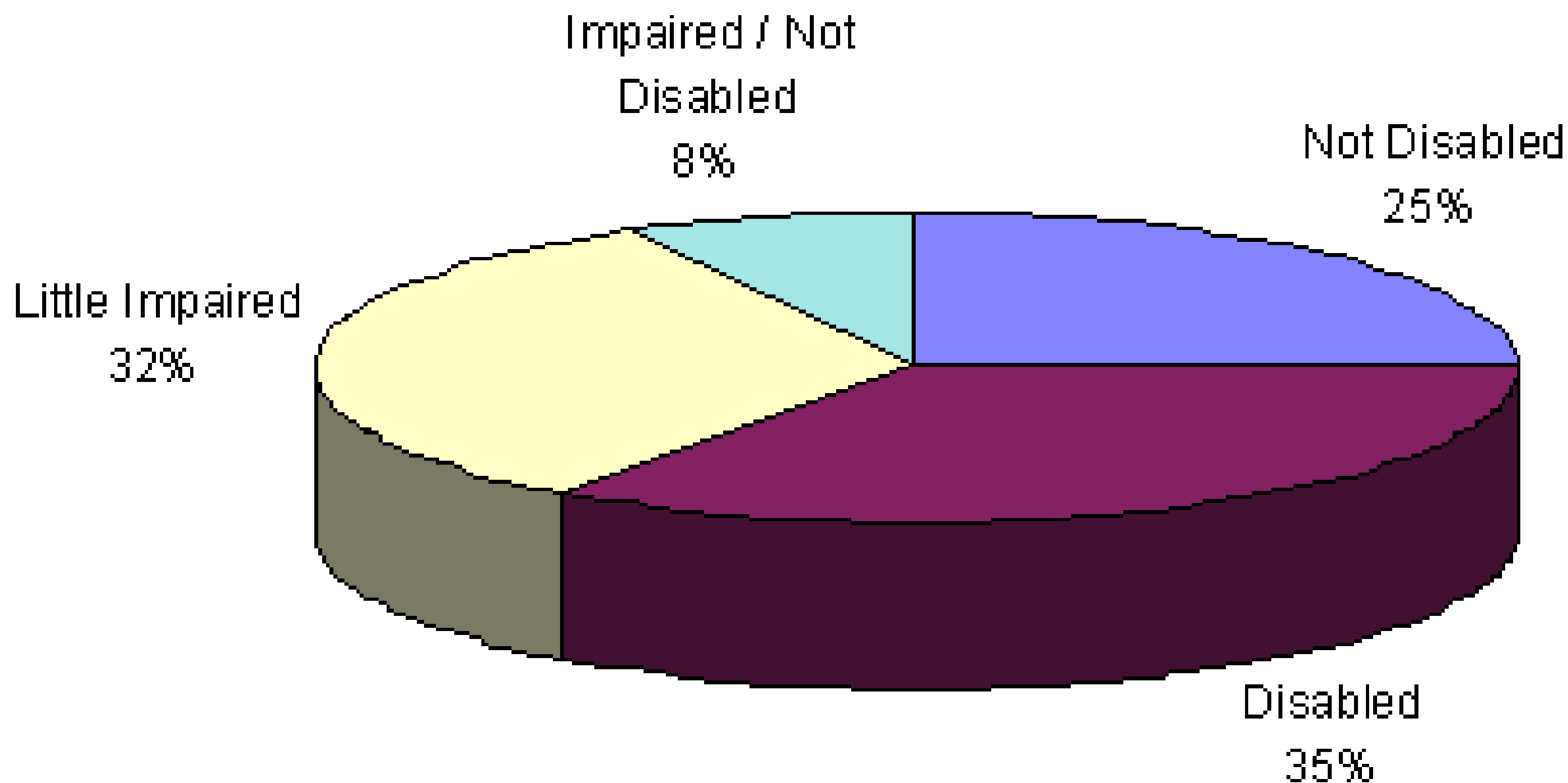
Source: Leray et al. *Mult Scler*, 2007.

Expanded Disability Status Scale (EDSS)

Severity	Description
Mild	EDSS score ≤ 3 1.0: No disability, minimal signs in one functional system. 1.5: No disability, minimal signs in more than one functional system. 2.0: Minimal disability in one functional system. 2.5: Mild disability in one functional system or minimal disability in two functional systems. 3.0: Moderate disability in one functional system, or mild disability in three or four functional systems. No impairment to walking.
Moderate	EDSS score 3.5 - 4.5 3.5: Moderate disability in one functional system and more than minimal disability in several others. No impairment to walking. 4.0: Significant disability, but self-sufficient and up and about some 12 hours a day. Able to walk without aid or rest for 500m. 4.5: Significant disability but up and about much of the day, able to work a full day, may otherwise have some limitation to full activity or require minimal assistance. Able to walk without aid or rest for 300m.
Severe	EDSS score 5.0 - 6.5 5.0: Disability severe enough to impair full daily activities and ability to work a full day without special provisions, Able to walk without aid or rest for 200m. 5.5: Disability severe enough to preclude full daily activities. Able to walk without aid or rest for 100m. 6.0: Requires a walking aid - cane, crutch etc., to walk about 100m without resting. 6.5: Requires two walking aids - pair of canes, crutches etc., to walk about 20m without resting.
Very Severe	EDSS score ≥ 7 7.0: Unable to walk beyond approximately 5m even with aid. Essentially restricted to wheelchair, though wheels self in standard wheelchair and transfers alone. Up and about in wheelchair some 12 hours a day. 7.5: Unable to take more than a few steps. Restricted to wheelchair and may need aid in transferring. Can wheel self but cannot carry on in standard wheelchair for a full day and may require a motorized wheelchair. 8.0: Essentially restricted to bed or chair or pushed in a wheelchair. May be out of bed itself for much of the day. Retains many self-care functions. Generally has effective use of the arms. 8.5: Essentially restricted to bed much of the day. Has some effective use of arms, retains some self-care functions. 9.0: Confined to bed. Can still communicate and eat. 9.5: Confined to bed and totally dependent. Unable to communicate effectively or eat/swallow.

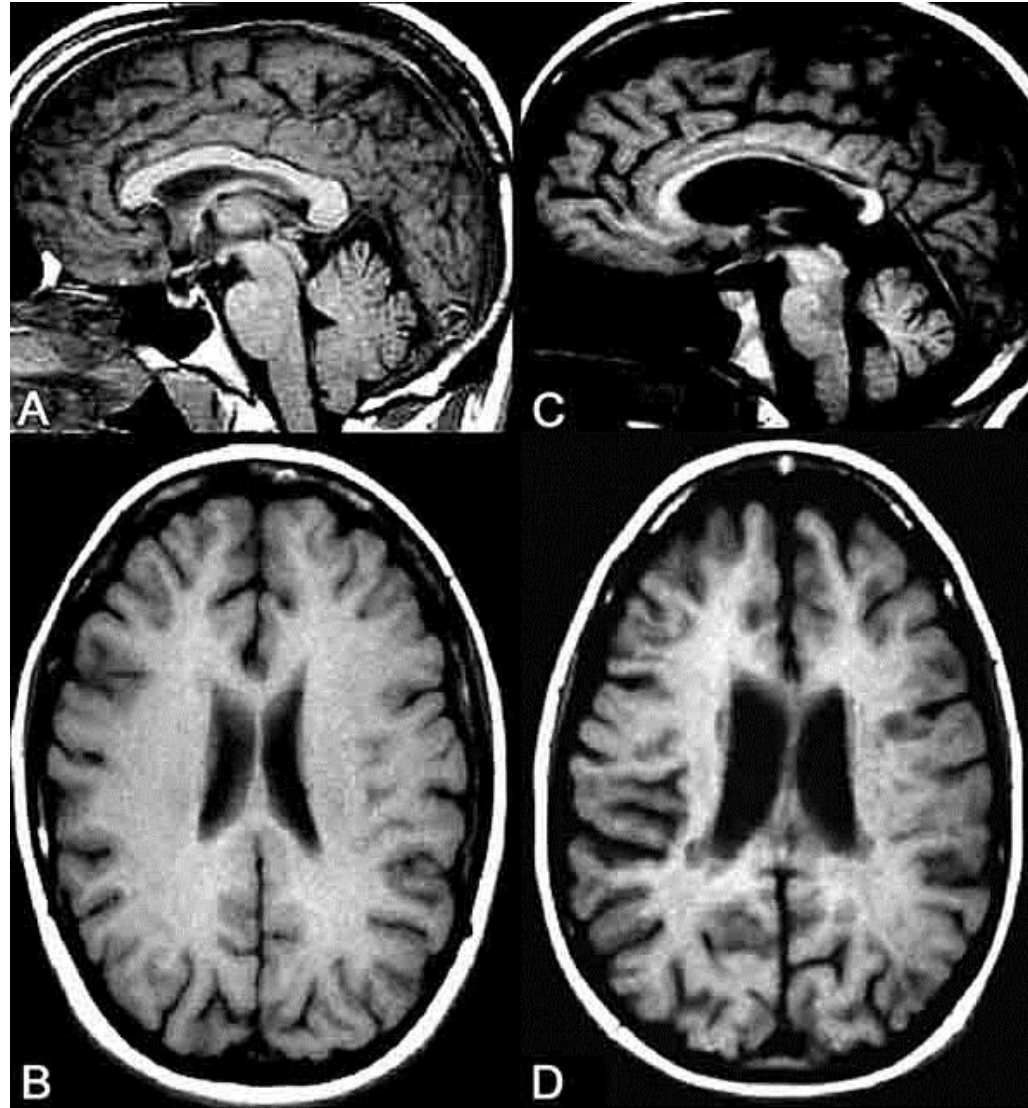
Source: www.mstrust.org.uk/atoz/edss.jsp

Morbidity – prognosis over 10-20 years

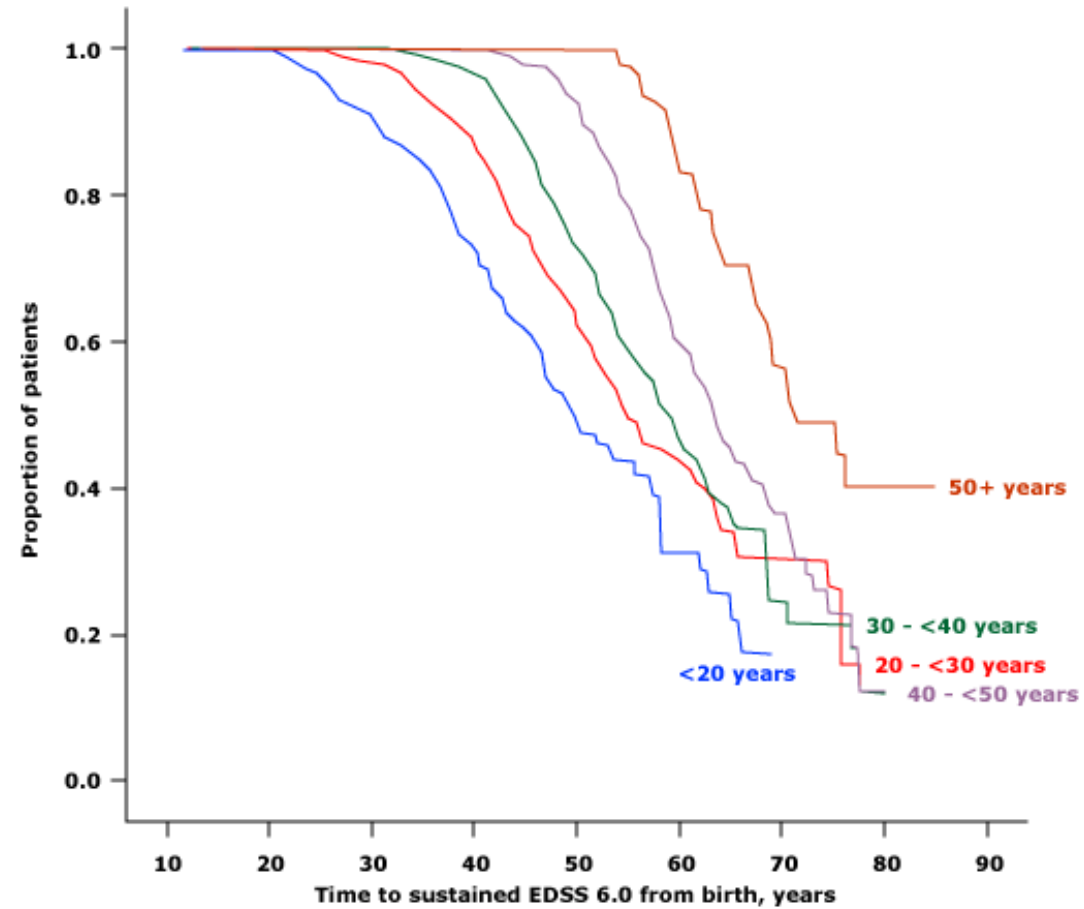


Source: <http://www.pneuro.com/publications/ms/index.html>

Brain atrophy in Multiple Sclerosis



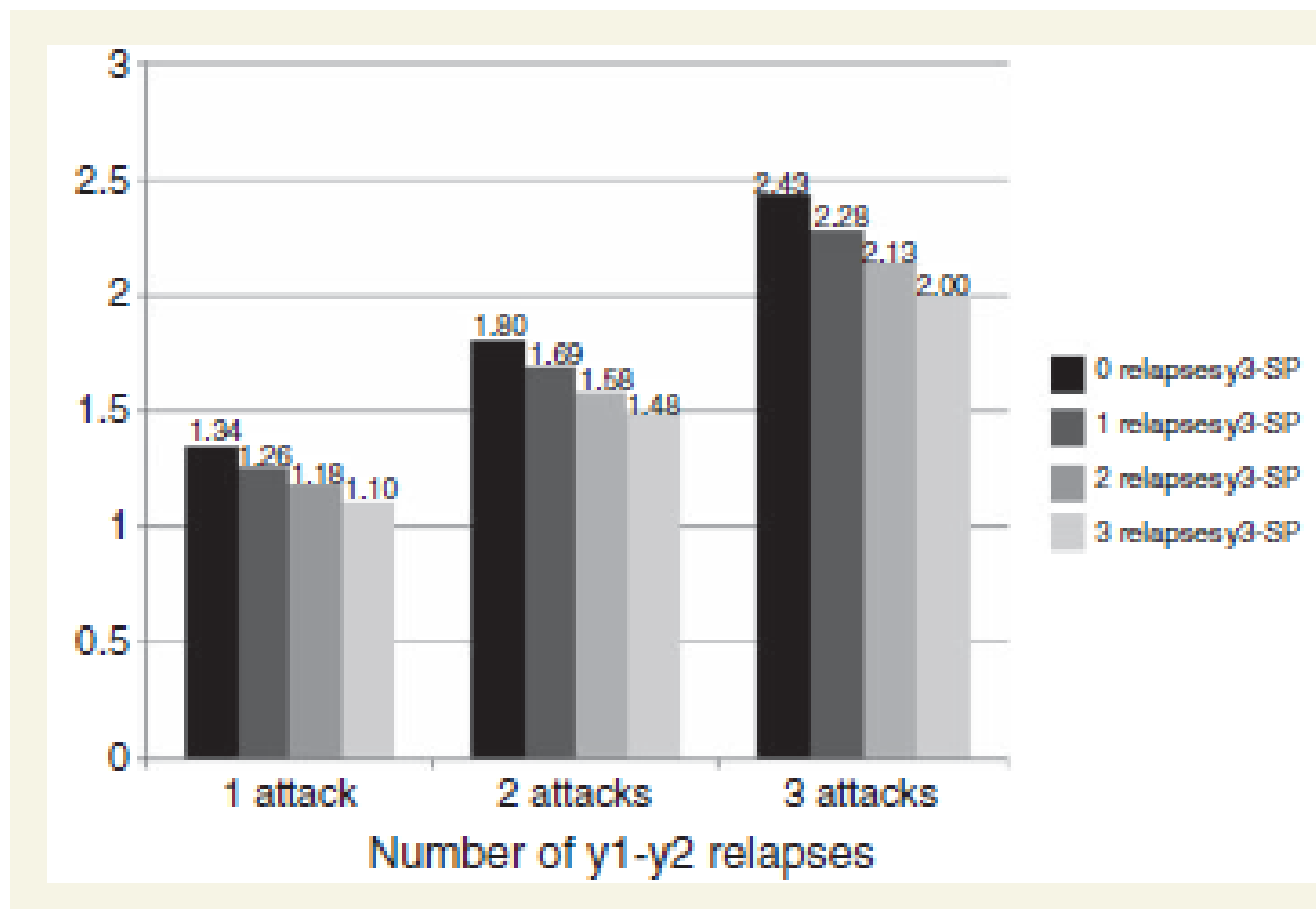
Time to sustained disability (use of a cane) with multiple sclerosis



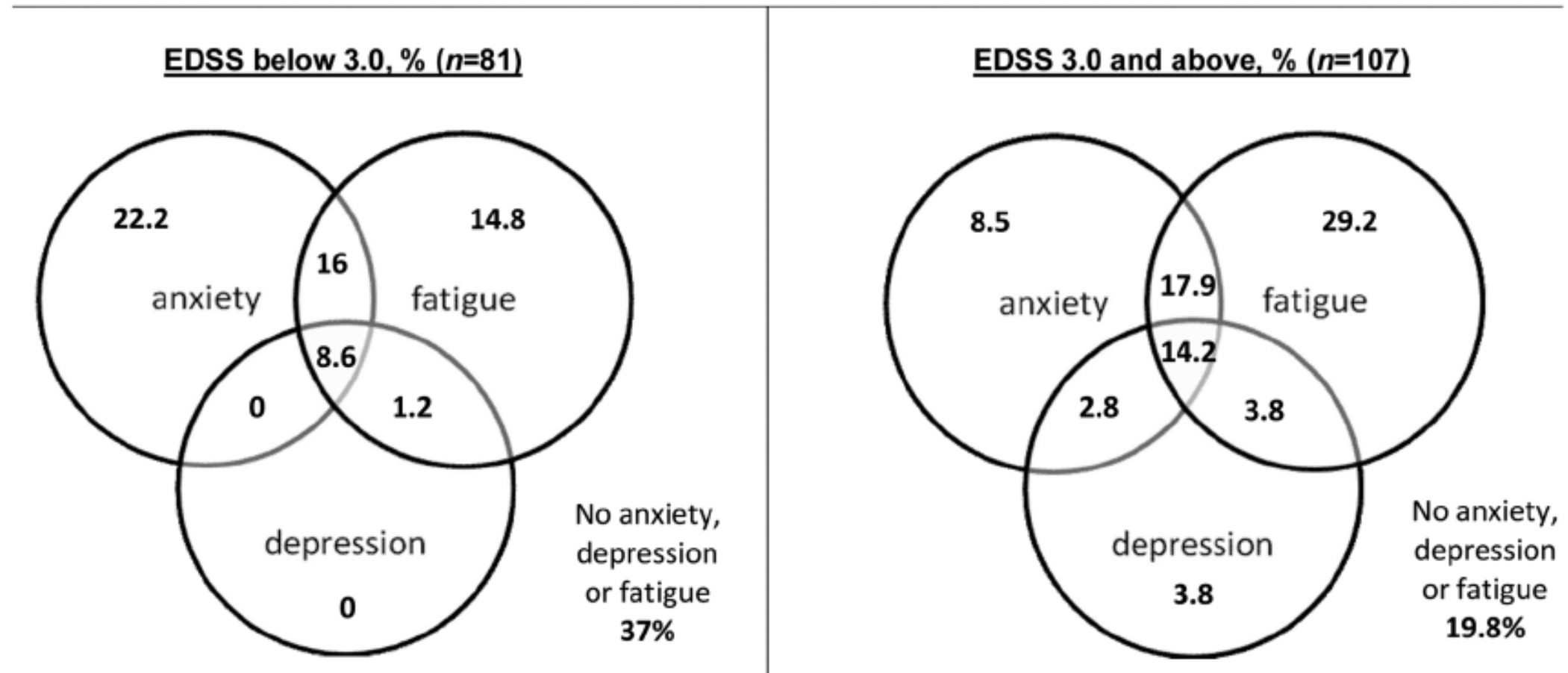
Kaplan-Meier survival curve of time to use of a cane (sustained Expanded Disability Status Scale score 6) from birth by age at onset of multiple sclerosis (n = 2319)

Reproduced with permission from: Tremlett H, Paty D, Devonshire V. Disability progression in multiple sclerosis is slower than previously reported. *Neurology* 2006; 66:172. Copyright © 2006 Lippincott Williams & Wilkins.

Prognosis according to number of relapses



Concurrent prevalence of anxiety, depression and fatigue by EDSS level.



ABI Recommended Definition for Multiple Sclerosis

Multiple sclerosis – *with persisting symptoms*

A definite diagnosis of Multiple Sclerosis by a Consultant Neurologist. There must be current clinical impairment of motor or sensory function, which must have persisted for a continuous period of at least 6 months.

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Claims considerations....

- A definite and secure diagnosis is required.
- ABI definition requires 'persisting symptoms'
 - should ideally be met... BUT, consider if...
 - Multiple episodes of symptoms that could be reasonably be linked.
 - Relatively severe level of impairment – would suggest at least EDSS 3...
 - Diagnosis is not under dispute.
- Fatigue can be considered as a functional impairment.
- Beware of non-disclosure!

RGIA

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